

Transparent Cross-Sector Emissions Scenarios Under Parameter Uncertainty

An auditable Python research prototype for scenario analysis, uncertainty, optimisation and physics-guided evaluation

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Abstract. Small organisations and researchers need emissions tools that are more inspectable than large integrated assessment systems and more rigorous than opaque spreadsheet totals. This paper presents an open Python prototype that represents greenhouse-gas estimates as traceable activity-by-emission-factor calculations, exposes scenario assumptions, propagates uncertainty, tests the stability of mitigation rankings and reports explicit interpretation boundaries. The software includes deterministic scenario comparison, one-at-a-time sensitivity, Monte Carlo analysis, a least-cost abatement optimiser, a synthetic physics-informed neural benchmark and a held-out comparison against 320 US EPA eGRID natural-gas plants. On that historical holdout, a physics-guided residual model reduced median absolute percentage error from 1.88% to 1.43% relative to a data-only ridge model, while RMSE was 0.06% worse. A paired 2,000-resample bootstrap estimated the median-error difference at -0.45 percentage points (95% interval: -0.64 to -0.25). These results support a limited claim about typical relative error within one 2018 dataset, not contemporary European validation, forecasting accuracy or realised emissions reduction. The contribution is therefore methodological: a compact, provenance-first workflow that makes assumptions, uncertainty, negative results and human interpretation requirements visible.

Keywords: greenhouse-gas modelling; scenario analysis; uncertainty quantification; Monte Carlo simulation; least-cost abatement; physics-informed machine learning; reproducible research

1. Introduction

Emissions scenarios are often interpreted more confidently than their data and assumptions allow. Activity data can be incomplete, emission factors can vary by geography and year, and apparently clear intervention rankings can reverse when uncertain inputs change. A useful decision-support system must therefore expose how every estimate was produced, what evidence supports each parameter and when the result is too unstable to defend.

This work asks four questions: which inputs dominate uncertainty; whether mitigation rankings remain stable across plausible ranges; how alternative system boundaries alter priorities; and whether provenance-first reporting can reduce unsupported interpretation of scenarios as forecasts. The resulting repository is an educational research prototype, not a regulatory inventory or operational control system.

2. System design

The core calculation is intentionally simple and auditable: emissions in kg CO₂e equal activity multiplied by an emission factor. Each activity record can retain its sector, unit, factor source, factor year and uncertainty. A scenario modifies named activity or factor values rather than hiding assumptions inside a fitted model. This design keeps the accounting boundary inspectable and allows uncertainty methods to operate on explicit parameters.

Component	Purpose	Evidence boundary
Scenario engine	Compare baseline and intervention assumptions	Conditional estimates, not forecasts
Sensitivity analysis	Rank local response to one-at-a-time changes	Diagnostic, not a confidence interval
Monte Carlo	Propagate transparent input ranges	Conditional on chosen distributions and independence
Least-cost optimiser	Meet an abatement target at minimum illustrative cost	Demonstration, not investment advice
Physics-guided benchmark	Compare data-only and constrained learning	Synthetic foundation plus one historical plant dataset

3. Evaluation studies

3.1 Hospitality water-energy case

A hospitality case study was motivated by firsthand observation of taps remaining on during repeated utensil-rinsing work. The repository converts editable assumptions about flow, operating time, avoidable use, hot-water share, temperature rise and heater efficiency into water, energy and emissions estimates. The observation motivates the question but does not establish a measured site baseline. Default values are illustrative and should be replaced by a documented measurement protocol before any operational claim is made.

3.2 Synthetic scientific-ML benchmark

A one-hidden-layer neural surrogate was trained twice from identical initial weights: once using only prediction error and once with an added penalty for disagreement with the physical relationship emissions = activity x emission factor. With seed 2026, 600 synthetic training samples, 15% observation noise and physics weight 1.5, the constrained model reduced interpolation RMSE from 9.69 to 6.22 kg CO₂e (35.8%) and extrapolation RMSE from 455.98 to 400.56 kg CO₂e (12.2%). Extrapolation error remained high, so the experiment demonstrates mechanics rather than reliable out-of-range prediction.

3.3 Held-out EPA eGRID plant comparison

The first real-data benchmark used a US EPA eGRID-derived 2018 natural-gas plant layer. Positive annual combustion heat input and reported CO₂ were retained. Plants whose ORISPL identifier was divisible by five formed a locked holdout and were never used for fitting. The final split contained 1,261 fitting records and 320 held-out plants. Target-derived emission-rate fields were excluded; reported annual CO₂ was used only as the evaluation target and within the fitting partition for fitted models.

Model	RMSE (short tons)	MAE (short tons)	Median absolute percentage error
Data-only ridge	32,058.72	9,694.49	1.88%
External-factor equation	32,541.30	9,920.03	1.58%
Physics-guided residual	32,079.37	9,579.33	1.43%

The physics-guided residual reduced median percentage error by 23.9% and MAE by 1.2% relative to the data-only ridge model. It did not reduce RMSE: squared error was 0.06% worse. Across 2,000 paired bootstrap resamples, the median-error difference was -0.45 percentage points with a 95% interval from -0.64 to -0.25; the guided model had lower absolute error on 57.5% of held-out plants. This interval quantifies sampling variation only.

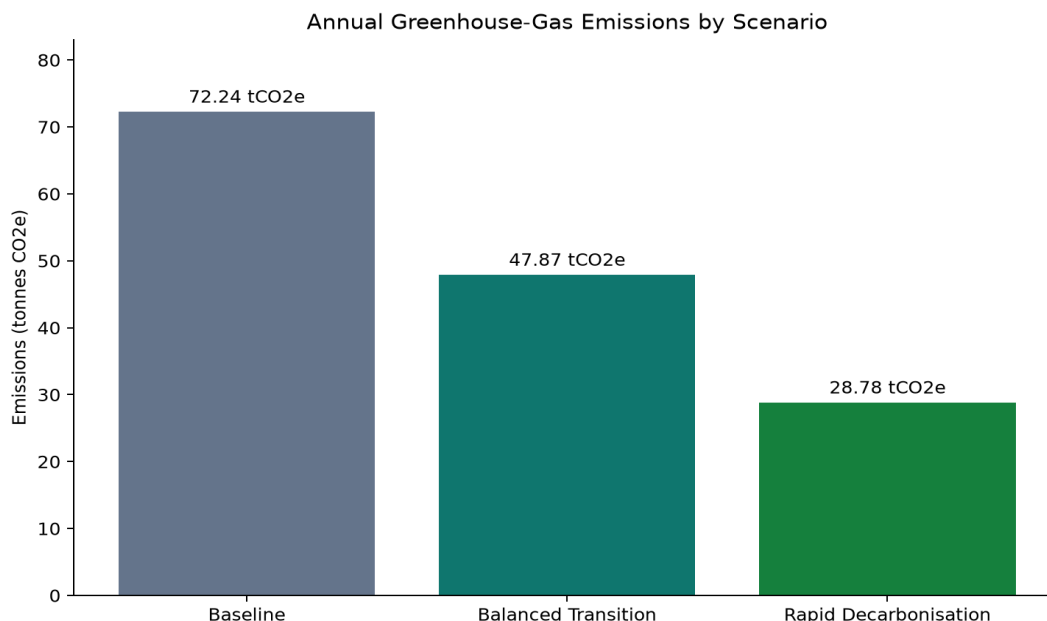


Figure 1. Example scenario totals generated by the open implementation. Values are conditional on the supplied activity data and emission factors.

3.4 Least-cost pathway and rank reversal

The optimiser represents candidate interventions as binary decisions and minimises total annualised cost subject to a minimum abatement target. A continuous relaxation is reported only as a lower bound. A companion rank-reversal analysis perturbs cost and abatement inputs and records how often each measure remains selected. On the illustrative eight-measure dataset, a 2,500 kg CO₂e/year target was met by five measures at approximately GBP 20/year net cost. Two net-saving measures were selected in every one of 1,000 perturbation trials, while the largest single-abatement option appeared in fewer than 2%. Because all cost and abatement values are placeholders, this result demonstrates the method only.

4. Reproducibility and governance

The repository provides command-line entry points, checked-in test fixtures, deterministic seeds, machine-readable outputs and automated tests across Python 3.10 and 3.12. Its central governance principle is that a result should travel with its evidence boundary: data source and year, unit and factor provenance, assumptions, uncertainty method, permitted interpretation and conditions requiring expert review.

- Open source under Apache License 2.0, with citation metadata in CITATION.cff.
- Thirty-nine automated tests cover calculations, validation rules, chart output, optimisation, sensitivity, Monte Carlo analysis and scientific-ML experiments.
- The real-data benchmark uses a deterministic holdout and excludes target-derived rate features.
- Negative results are retained, including the slight RMSE deterioration and high extrapolation error.

5. Limitations

- The real-data comparison is limited to unseen natural-gas plants in one US historical year; it is not contemporary, European, temporal or cross-sector validation.
- Both activity and target originate in the eGRID data product, although the fixed physical factor is external to the target.
- The hospitality operating assumptions and abatement costs are illustrative rather than audited measurements or vendor quotations.
- Monte Carlo inputs are presently sampled independently; real operating and price variables can be correlated.

- The physics-informed neural experiment uses a simple equation and synthetic data and does not establish competence with neural operators or production scientific ML.
- No result demonstrates causal emissions savings, regulatory compliance, forecast skill or suitability for capital allocation.

6. Next study phase

The next defensible extension is a temporal and jurisdictional replication: import a later eGRID year without changing the locked model, then reproduce the benchmark on EU ETS installation data with explicit unit and boundary reconciliation. The scenario layer should add a versioned factor registry, correlated uncertainty, variance decomposition and independently prepared inventory reconciliation. The hospitality case requires direct measurement of flow, duration, temperature and service constraints before any intervention study.

7. Conclusion

This prototype shows that compact emissions software can combine transparent accounting, uncertainty analysis, optimisation and physics-guided learning without hiding weak or negative results. The present evidence supports reproducibility and a limited historical holdout result. It does not support operational, forecasting or policy claims. That distinction is the project's main contribution: modelling conclusions should remain attached to their provenance, uncertainty and permitted interpretation.

References and resources

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Research artefact statement

Software, documentation, test fixtures and reproducibility commands are available in the public repository. The project is educational research. Users should consult qualified carbon-accounting and domain experts before applying any result to reporting, procurement or operational decisions.